

Hybrid Semantic Question Answering System Over Academic Data

1. Project Title

A Hybrid Semantic Question Answering System over MOOC Catalogs: Combining Ontology-Based Retrieval with Retrieval-Augmented Generation (RAG)

2. Abstract / Problem Definition

Traditional systems in online course repositories usually rely on standard keyword matching, which fails to capture the conceptual meaning behind a user's query. To solve this, recent artificial intelligence architecture focuses on Question Answering (QA) using vector-based Retrieval-Augmented Generation (RAG). While RAG excels at understanding open-ended semantic meaning, it lacks structural awareness and logical reasoning. This means it cannot guarantee completely accurate relational facts, such as matching exact course levels or specific university constraints. Conversely, purely structured approaches like Ontologies offer perfect factual precision but cannot easily read open-ended, unstructured text paragraphs.

This project implements a Neuro-Symbolic Hybrid Question Answering System over a massive open online course (MOOC) repository. By combining an Ontology module (Declarative Semantics) with a dense Vector Retrieval module (Distributional Semantics), our architecture bridges the gap between structured relational precision and unstructured contextual flexibility. The outputs of both systems pass through a Result Fusion Layer to deliver logically accurate, highly relevant, and contextually rich answers to complex user queries.

3. Dataset & Feature Engineering

We utilize the **Coursera Course Dataset** sourced from HuggingFace (azrai99/coursera-course-dataset), which contains **6.5k unique course profiles**. To maintain strict consistency across both modules, the data columns are partitioned as follows:

- **Structured Data (Ontology Pipeline):** We extract relational metadata features including title, Instructor, Organization (University), and Level to build our deterministic graph.
- **Unstructured Data (RAG Pipeline):** We extract the text blocks from the Description and Modules/Courses columns to build our dense semantic vector index.

4. Methodology & Architecture Components

The system processes user queries through two parallel pipelines before merging the results:

1. **Declarative Semantics Layer:** Using the Python library `rdflib`, we construct an in-memory RDF Triple Store governed by a custom, lightweight schema. We define core classes (`ex:Course`, `ex:Instructor`, `ex:Organization`) and explicit predicates (`ex:taughtBy`, `ex:offeredBy`). Exact factual retrieval is executed via **SPARQL queries**.
2. **Distributional Semantics Layer:** The unstructured course descriptions are embedded into a vector space using a pre-trained Transformer model (`sentence-transformers/all-MiniLM-L6-v2`). Semantic similarity search is executed using a **FAISS (Facebook AI Similarity Search)** index.
3. **Result Fusion Layer:** When a query is received, the RAG module finds the most contextually relevant courses via FAISS. The IDs of these matches are then dynamically passed into a SPARQL query. This allows the Ontology module to enforce strict structural constraints (such as verifying the correct professor or difficulty level), blending conceptual search with guaranteed factual truth.

5. Evaluation Plan (Quantitative & Qualitative)

In line with the course guidelines, our evaluation focuses on understanding the strengths and limitations of this hybrid approach compared to standalone baseline models:

- **Quantitative Metrics:** We will create a test benchmark of 20 to 30 complex, multi-hop queries. We will evaluate our retrieval accuracy using standard information retrieval metrics: **Precision**, **Recall**, and **Mean Reciprocal Rank (MRR)**.
- **Qualitative Error Analysis:** We will systematically analyze cases where the system returns suboptimal results. Failures will be categorized into specific bottlenecks (e.g., missing triples in the graph, semantic drift in the vector embeddings, or scoring misalignments in the fusion layer) to reflect on the structural limits of our pipeline.

6. Reference Papers

1. **Gupta, S., Ranjan, R., & Singh, S. N. (2024).** *"A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions."* arXiv preprint. [Link](#)
2. **Pan, S., et al. (2024).** *"Unifying Large Language Models and Knowledge Graphs: A Roadmap."* IEEE Transactions on Knowledge and Data Engineering. [Link](#)
3. **Gajderowicz, B., Bhardwaj, A., & Fox, M. S. (2025).** *"RAG and Ontologies for Information Retrieval: A Literature Review."* University of Toronto. [Link](#)
4. **Yang, X., et al. (2024).** *"CRAG - Comprehensive RAG Benchmark."* Advances in Neural Information Retrieval Systems (NeurIPS Datasets and Benchmarks Track). [Link](#)
5. **Galla, D., et al. (2024).** *"CoURAGE: A Framework to Evaluate RAG Systems."* Natural Language Processing and Information Systems (NLDB 2024), Springer. [Link](#)